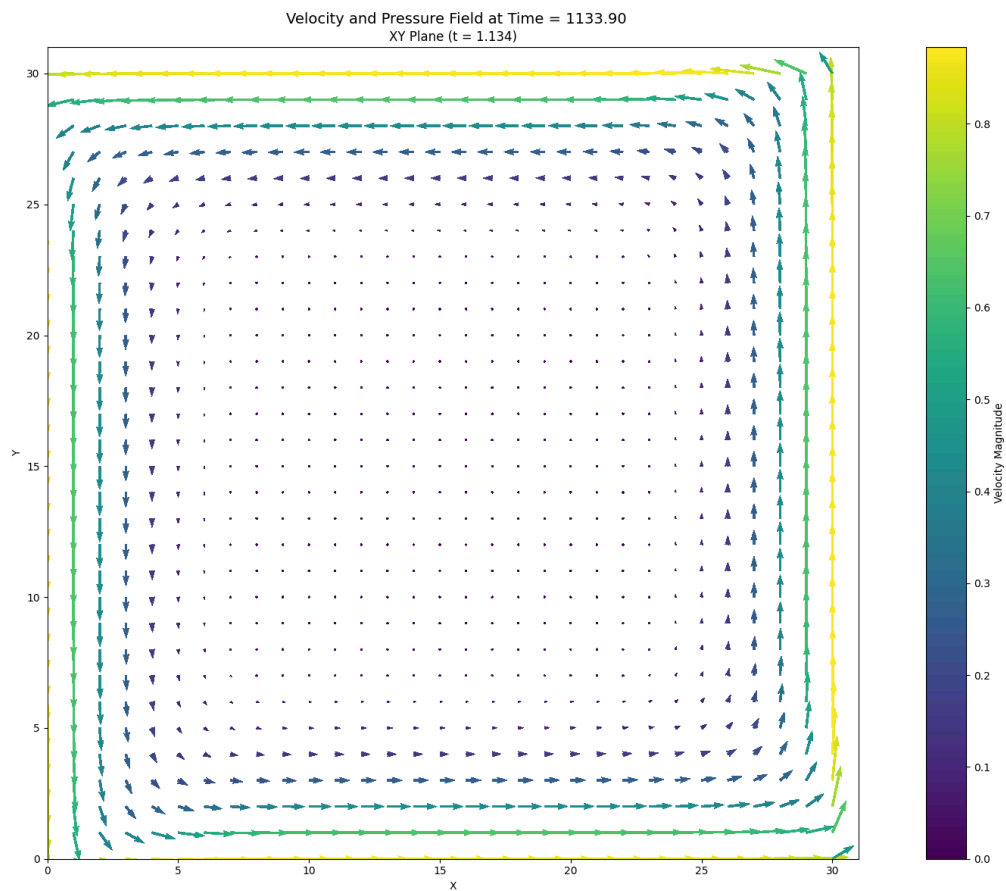


Experimental Report

Case: Lid Cavity with 4 Moving Walls

Description of Flow: This experiment studies a two-dimensional cavity flow in which all four walls move at prescribed speeds, setting the fluid into a strong recirculating motion. The moving boundaries inject momentum into the fluid and generate a nonlinear vortex structure with both shear-driven and pressure-driven effects. The objective is to identify the governing streamwise momentum equation directly from the evolving velocity and pressure fields.

Numerical identification of the streamwise momentum equation from saved velocity and pressure fields.



1. Governing Equation Verification

Reference equation

$$u_t = -1.00000000e+00*uu_x - 1.00000000e+00*vu_y + 1.00000000e-02*u_{xx} + 1.00000000e-02*u_{yy} - 1.00000000e-03*p_x$$

$$v_t = -1.00000000e+00*uv_x - 1.00000000e+00*vv_y + 1.00000000e-02*v_{xx} + 1.00000000e-02*v_{yy} - 1.00000000e-03*p_y$$

Learned equation

$$u_t = -9.26171495e-01uu_x - 5.86478455e-01vu_y + 1.17845623e-02u_{xx} + 7.83468588e-03u_{yy} - 6.20912385e-04p_x$$
$$v_t = -8.34780067e-01uv_x - 1.20858250e+00vv_y + 8.07767889e-03v_{xx} + 7.56703159e-03v_{yy} - 1.01070735e-03p_y$$

Term	Reference coefficient	Learned coefficient	Absolute error
uu_x	-1.00000000e+00	-9.26171495e-01	7.38285050e-02
vu_y	-1.00000000e+00	-5.86478455e-01	4.13521545e-01
u_xx	+1.00000000e-02	+1.17845623e-02	1.78456232e-03
u_yy	+1.00000000e-02	+7.83468588e-03	2.16531412e-03
p_x	-1.00000000e-03	-6.20912385e-04	3.79087615e-04

Term	Reference coefficient	Learned coefficient	Absolute error
uv_x	-1.00000000e+00	-8.34780067e-01	1.65219933e-01
vv_y	-1.00000000e+00	-1.20858250e+00	2.08582497e-01
v_xx	+1.00000000e-02	+8.07767889e-03	1.92232111e-03
v_yy	+1.00000000e-02	+7.56703159e-03	2.43296841e-03
p_y	-1.00000000e-03	-1.01070735e-03	1.07073471e-05

2. Simulation Parameters

Quantity	Symbol	Value	Unit
Fluid density	rho	1.00000000e+03	kg m ⁻³
Dynamic viscosity	viscosity	1.00000000e+01	Pa s
Kinematic viscosity	nu	1.00000000e-02	m ² s ⁻¹
Characteristic velocity	u_max	1.00000000e+00	m s ⁻¹
Characteristic length	D	1.00000000e+00	m
Reynolds number	Re	1.00000000e+02	dimensionless
Body force	G	9.81000000e+00	m s ⁻²
Grid points in x	Nx	3.10000000e+01	cells
Grid points in y	Ny	3.10000000e+01	cells
Grid spacing in x	dx	3.22580645e-02	m
Grid spacing in y	dy	3.22580645e-02	m
Time step	dt	1.00000000e-04	s
Numerical threshold	epsilon	7.00000000e-11	-

3. Data and Pre-processing

Quantity	Value
Loaded field shape (t, x, y)	(9851, 31, 30)
Regression field shape (t, x, y)	(7851, 28, 29)
Regression samples	6,375,012
Dictionary terms	5
First saved time index	150
Temporal crop at each boundary	2000 layers
x-boundary crop	1 points
y-boundary crop	1 points

4. STRidge Selection and Error Metrics

Metric	Value
L0 penalty	1.00000000e-06
Regularisation parameter, λ	1.00000000e-09

Metric	Value
Selected tolerance	5.87801607e-04
Training RMSE	9.15047149e-02
Validation RMSE	9.39825989e-02

Metric	Value
Selected tolerance	9.42668455e-04
Training RMSE	8.05213761e-02
Validation RMSE	8.02861086e-02